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# Palaeomagnetism and Rb-Sr ages of the Ratagan and Comrie intrusions

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**Summary**. Study of the palaeomagnetism of two complexes from the Newer Granite Suite in Scotland, at Ratagan (NW Highlands) and Comrie (central Highlands), reveals the variable nature of the natural remanence encountered in granodioritic intrusions and the surrounding metamorphic country rock. Forty-eight specimens from Ratagan, dated at 415 ± 5 Ma, gave a mean direction: $D = 8°$, $I = -32°$, and a palaeomagnetic south pole: 15°S, 346°E ($\delta p = 5°$, $\delta m = 9°$). Twenty-eight specimens from Comrie, dated at 408 ± 5 Ma, gave a mean direction: $D = 75°$, $I = -30°$, and a palaeomagnetic south pole: 6°S, 287°E ($\delta p = 4°$, $\delta m = 7°$). These results have been compared with the established apparent polar wander path (APWP) for Britain. The Ratagan pole improves the reliability of the APWP but doubt remains as to whether the primary magnetization from Comrie represents a true late Silurian direction or whether it has been affected by post-cooling rotation, possibly associated with the nearby Highland Boundary Fault.

**Key words:** palaeomagnetism, Palaeozoic intrusions, Rb-Sr ages, Scotland

## 1 Introduction

The newer igneous rocks in the Scottish Highland region which post-date the main Grampian metamorphism and tectonism range in composition from basic to granitic. Two members of this suite were chosen for palaeomagnetic study in an attempt to refine the apparent polar wander path for Scotland in the Palaeozoic.

This paper describes the results of the detailed palaeomagnetic analysis which has been carried out. Thermal and alternating field (af) demagnetization was supplemented by rock magnetic experiments on a limited selection of specimens from each sampling site to help characterize the remanence carriers. These consisted of thin-section observations and the measurement of thermomagnetic and hysteresis properties. Each complex provides an example of the widespread variability in remanence behaviour which may be encountered in a single intrusion.

For an accurate interpretation of palaeomagnetic data from any formation, an estimate of the age of the rock is required. Brown, Miller & Grasty (1968) give a poorly defined K-Ar

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isotopic age for the Ratagan complex and no data are reported in the literature for the Comrie intrusion. Rb-Sr isotope analyses were carried out on samples from both these intrusions in order to make some reliable age determinations.

## 2 Method

Processing methods adopted were the same for all the palaeomagnetic samples. Individual specimens from each site were thermally demagnetized in steps of 50°C or smaller intervals depending on the behaviour of the rock. Another specimen from each site was af demagnetized in steps of 5–10 mT. After a definite pattern of behaviour had emerged for each site, the rate of processing was increased by using fewer, appropriately chosen temperature and field intervals.

Components of magnetization together with their blocking temperature ranges were estimated with the aid of orthogonal projections of the demagnetization data. As a result of minimizing the number of field/temperature intervals, the remanences removed at low steps, which were considered to have no ancient geological significance, were often defined by only two points on the orthogonal projections.

Generally, orthogonal projections of demagnetization data reveal straight-line segments which are associated with single components of NRM. In any assessment of such vector plots, some decision must be made as to which points should be included in the definition of a particular component. In this study, it was found that minimization of the 95 per cent confidence angle on vector differences ($\Delta_{\gamma 95}$ of Briden & Arthur 1981) resulted in a tighter grouping of the removed vectors. Therefore, this was used as a guide to obtaining 'best-estimates' of the direction of magnetization components (cf. Turnell & Briden 1983).

Listings of all the components from each specimen used in this study, including blocking temperature/coercivity ranges and the error parameter, are available from the author.

## 3 The Ratagan Complex

### 3.1 GEOLOGY AND SAMPLING

The Ratagan Complex is a composite intrusion situated approximately 10 km east of Glenelg in Inverness-shire, east of the Moine thrust. To the north-east the igneous rocks intrude Lewisian paragneisses and elsewhere the country rock comprises Moinan gneisses interrupted by pelitic bands. Immediately east of the pluton, the country rocks are crushed and shattered along faults of the Strathconon–Ben Sgriol system (Nicholls 1950). Parallel, SW–NE fractures cut the igneous rocks but have little effect on the igneous boundaries.

Fig. 1 is a sketch map of the intrusion showing the four main phases. From west to east these are classified as granite, diorite, syenitic hybrids and adamellite (Nicholls 1950). Each phase itself shows considerable variation in both composition and grain size and, taken as a whole, the body has an unusually high alkali element content compared with intrusions in the central and eastern Highlands. Nevertheless, Ratagan has been classified as one of the Newer Granites of Scotland. Nicholls (1950) concluded that the body is derived from a parental dioritic magma through differentiation and hybridization and more recent major and trace element analyses (Turnell 1982) have revealed low Nb/Y ratios relative to within-plate alkali basalts suggesting that they are not intra-plate.

There is little evidence on the form of the intrusion and, because it post-dates the main cleavage seen in the country rock, no structural correction has been applied to the palaeomagnetic directions obtained.

A total of 18 sites were drilled in the intrusion itself, in road cuttings, the river bed and along forestry tracks. The distribution of sites is shown on the sketch map (Fig. 1). Nine

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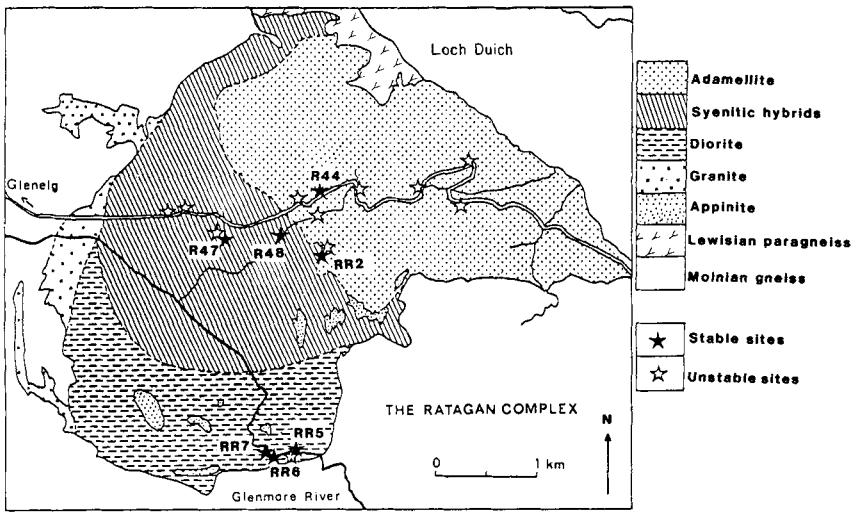



**Figure 1.** Sketch map of the Ratagan complex showing the distribution of sampling sites.

sites from the adjacent country rock west of Ratagan formed an E–W traverse. Immediately adjacent to the syenite the cores were from Moinan gneiss but the more westerly sites were drilled in the variable rock types of the Lewisian inlier.

### 3.2 RESULTS

For most sites, natural remanent magnetizations (NRMs) clustered close to the direction of the Earth's present field. Values of Koenigsberger ratios (*Qn*) calculated for samples from the adamellite, syenite and diorite were below 0.5, reflecting the presence of coarse-grained magnetite. For specimens from the diorite phase *Qn* values were higher, suggesting single domain (sd) or pseudo-single domain (psd) magnetite.

Af demagnetization of a large proportion of pilot samples failed to isolate any consistent direction of magnetization which might be geologically ancient. This unstable behaviour is attributed to multi-domain (md) magnetite which is affected by stray fields during measurement or in the demagnetizer. Typical demagnetization results are shown in Fig. 2(a). Stable specimens from the diorite had coercivities in the range 100–140 mT corresponding to sd or psd magnetite. Stable specimens from the syenite contain haematite with coercivities exceeding 200mT. The corresponding stereograms show southerly downward or north-easterly upward directions for the diorites and exclusively north-easterly upward directions for the syenites.

Thermal demagnetization revealed that most specimens contained some proportion of low blocking temperature (Tb) material. Stereograms and orthogonal projections showing the directional changes of the NRM during demagnetization are shown in Fig. 2(b). The stable specimens were classified into three groups: (1) Most syenite specimens appeared to contain only one (ancient) component of remanence (Tb > 580°C) in a northerly, upward direction, presumably carried by haematite, e.g. R48-3. (2) The dioritic specimens contained two (ancient) remanence components, one in a northerly upward direction, and the other in a southerly downward direction, e.g. RR6-2, RR7-7; both had Tb < 580°C and are presumably carried by magnetite. Specimen RR5-2, of diorite composition, was atypical in revealing only a southerly component. A few samples were found, upon heating to reveal, in

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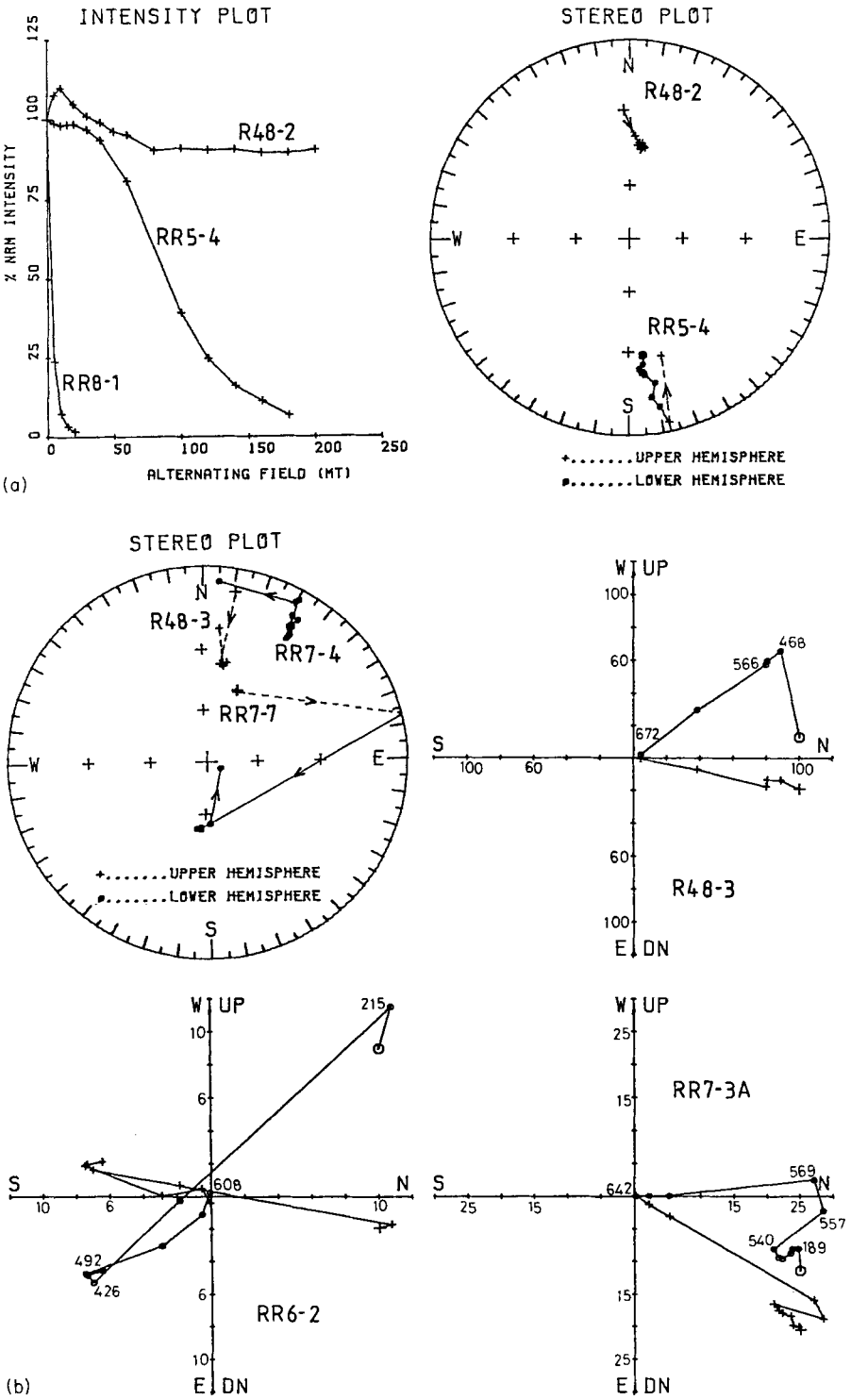



**Figure 2.** (a) Behaviour of specimens from the dioritic (RR5-4) and syenitic (R48-2) phases in Ratagan upon af demagnetization. RR8-1 is typical of specimens showing unstable behaviour. (b) Examples from each of the three classes of NRM identified after thermal demagnetization, as described in the text. On the orthogonal projections circles represent points on the vertical plane and crosses represent points on the horizontal plane. Temperatures, in °C, mark the points chosen for calculating the removed vectors. The axes are marked in units of $\times 10^{-3}$ A m$^{-1}$.

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addition to the group 2 components, a haematite remanence (Tb *c.* 570–640°C) showing a similar direction to the lower temperature magnetite component. In these specimens, the the higher temperature magnetite component was only observed in a small temperature interval, *c.* 20°C, and was not completely separated from the haematite over-print, e.g. RR7-3A (Fig. 2b). Two specimens from R44 were also found to contain magnetizations carried by both haematite (northerly upward) and magnetite (southerly downward).

### 3.3 MAGNETIC PROPERTIES

Two types of thermomagnetic behaviour were observed as shown in Fig. 3(a): (a) the convex curves are typical of specimens containing nearly pure sd magnetite which dominates the results even for specimens with a high proportion of primary remanence carried by haematite (e.g. R44, R48); (b) the concave curves indicate the presence of superparamagnetic grains (Radhakrishnamurty 1973) although this does not appear to affect the stability of the specimen on thermal demagnetization in every case (e.g. RR7-3). Possibly the relaxation time is comparable to the laboratory measurement time.

Most specimens gave saturation isothermal remanent magnetization (IRM) curves (Fig. 3b) with $B_{cr}$ (the coercivity of maximum remanence) between 20 and 50 mT, typical of nearly pure magnetite. Haematite bearing specimens, e.g. R48-2, failed to achieve saturation in field strengths of around 0.7 T. Unstable specimens tended towards much higher saturation magnetizations, indicating a higher concentration of coarse, probably md magnetite.

Thin-section studies revealed that, while all the material shows some degree of hydrothermal alteration in the form of brown-red iron staining and sericitization of feldspars, this was seen to be more extensive in the syenitic rocks. In general, the highest proportion of coarse-grained magnetite was observed in specimens from sites which showed the poorest magnetic stability. The total ferromagnetic content observed in the stable dioritic rocks was small compared with the stable syenitic specimens, reflecting their relatively lower NRM intensities ($< 20 \times 10^{-3}\,\mathrm{Am}^{-1}$). Observations suggest that the source of the stable haematite

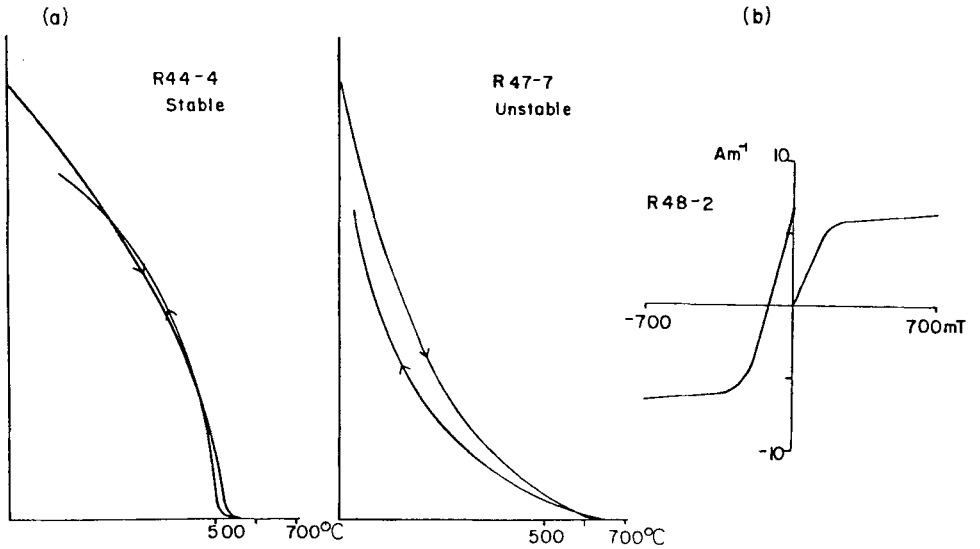



**Figure 3.** Typical (a) thermomagnetic and (b) saturation IRM curves from Ratagan samples. The vertical scale for the thermomagnetic curves, $J_S$, is arbitrary.

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palaeomagnetic component is either (a) hydrothermal alteration products or (b) exsolution lamellae. It is not possible to determine whether any phase contributes purely thermoremanent magnetization (TRM) or chemical remanent magnetization (CRM).

### 3.4 INTERPRETATION

Confidence circles associated with the two pole positions derived independently from the normal (N) and reversed (R) directions in each site show that they are not significantly different from normal and reversed indicators of the same magnetic field direction and are compatible with other known lower Palaeozoic poles from Scotland (e.g. Piper 1979). Therefore, all the data have been combined to produce a single pole position.

The simplest explanation for the origin of the magnetization is that it is a TRM recorded during the cooling of the intrusion through regional uplift. Considering the magnetite remanence, the consistently higher blocking temperature of the reversed component suggests that it is older than the normal component. Such an argument would not hold if either of the magnetite components is not a pure partial thermoremanent magnetization (PTRM). The haematite remanence, which is of normal polarity, could be a thermochemical remanence (TCRM) formed during haematite growth during the later stages of cooling when the lower Tb magnetites were acquiring their PTRM. It is unlikely to be much younger because of the agreement with the lower Tb magnetite component.

### 3.5 COUNTRY ROCKS

Fig. 4 shows the thermal demagnetization behaviour of specimens from the metamorphic rocks west of Ratagan. Reasonably well-grouped remanences were found within the Lewisian in amphibolitic material (site R55) identified as part of the pre- or syn-tectonic amphibolite suite (Smith 1979), in high-grade gneiss, permeated with granitic veins (sites R50 and R52) and in reddish Moinian quartzite (site R54). Other samples, from lower-grade Lewisian psammite and some Moinian gneiss were found to be weak and unstable.

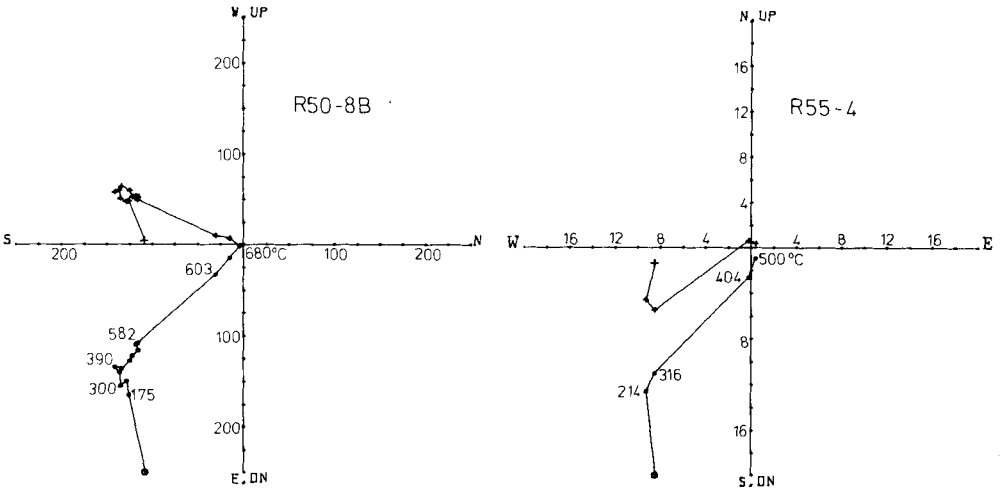



**Figure 4.** Thermal demagnetization of metamorphic rocks west of Ratagan. Both examples reveal component B, carried by magnetite and haematite in R50-8B and by magnetite only in R55-4. Circles represent points on the vertical plane and crosses represent points on the horizontal plane. The axes are marked in units of $\times 10^{-3}$ A m$^{-1}$.

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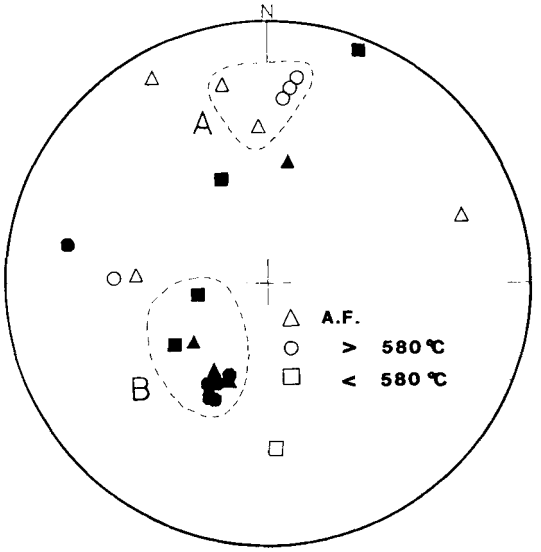



**Figure 5.** Stereographic plot for the metamorphic rocks west of Ratagan illustrating the removed vectors from af and thermal demagnetization. Solid symbols represent lower hemisphere directions, open symbols represent upper hemisphere directions.

Apart from a few scattered points, corresponding to poorly defined, contaminated or anomalous magnetizations, the data define two palaeomagnetic directions, A and B (Fig. 5). Blocking temperatures and coercivities indicate haematite remanence carriers for A and both magnetite and haematite carriers for B. The A component is similar to that in the Ratagan complex; the B direction is better defined from a larger population. It is significantly different from the Ratagan direction and by comparison with the established apparent polar wander path (APWP) for this period (e.g. Turnell & Briden 1983) must be younger. However, the source of this component is unclear.

## 4 The Comrie intrusion

### 4.1 GEOLOGY AND SAMPLING

A simplified geological map of the Comrie area is shown in Fig. 6. The major part of the body is composed of diorites which are themselves intruded by a plug of medium to fine-grained pink granite. The country rock comprises hornfelsed slate and greywacke of the Dalradian series. Because of the observed variation in both composition and grain size, sampling sites were restricted to the finer-grained, most basic phases, which are most suitable for palaeomagnetic work. Some cores were also drilled in the country rock up to 6 km from the diorite body making a total of 17 sampling sites in the area. These are shown on the sketch map (Fig. 6).

### 4.2 RESULTS

NRMs (except the country grit) measured between 50 and $500 \times 10^{-3}$ $Am^{-1}$ and the directions of magnetization grouped around the Earth's present field. For most specimens, Koenigsberger ratios, $Qn$, were close to 0.2 indicating a high proportion of md magnetite.

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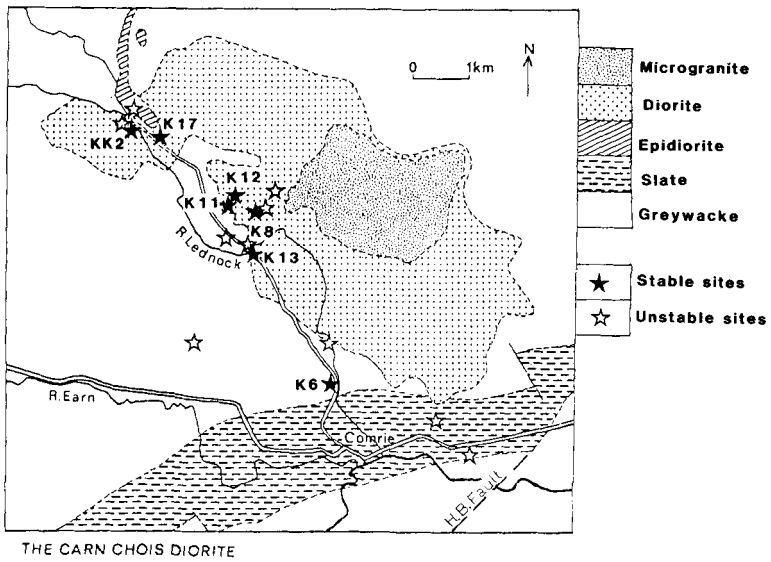



**Figure 6.** Sketch map of the Charn Chois diorite complex at Comrie showing the distribution of sampling sites.

Upon af demagnetization, single magnetization components, with coercivities in the range 40–100 mT, characteristic of sd magnetite, were isolated from nearly all the fine-grained diorites. Most of the directions were ENE, upward. In examples from K12 and K13, however, the magnetization was shallower or even slightly downward with a more northerly declination. Af treatment of one specimen, K17-6A, revealed a southerly downward component. Examples of some of the diorite specimens are shown in Fig. 7(a). In every case, when the intensity had fallen to about 10 per cent of the NRM, *c.* $10 \times 10^{-3}$ A m$^{-1}$ at field strengths of between 30 mT (for K13-5A) and 100 mT (for K11-1A) measurements of the remanence were no longer repeatable. All the specimens from the sites in the granite, granodiorite and Dalradian country rock, outside the highly metamorphosed contact zone, proved to be unstable on af treatment and similar behaviour was shown by some of the hornfels specimens.

The intensity demagnetization curves (Fig. 7a) confirm that all the specimens, from both stable and unstable sites, contained a high proportion of low coercivity material. Some of this may have a low blocking temperature and short relaxation time and thus contribute to the residual noise after af cleaning. Such components could also explain the few anomalous directions observed.

Classification into stable and unstable groups on the basis of thermal demagnetization behaviour produced the same divisions as for the af results. The stability of only four out of eight specimens from site KK2 and the difference in components isolated from individual cores from K17 demonstrate the considerable within site variation in magnetic remanence. Examples of demagnetization behaviour are shown in Fig. 7(b).

Component A, mean $D = 75°$, $I = -30°$, $\alpha_{95} = 6.3°$ was identified as the characteristic direction, corresponding to the original cooling of the formation. Some specimens, from sites K13, K12 and K8 appear to have primary components with shallower or positive inclinations and north-easterly directions. However, the spread of these components approximates to a great circle between A and the present-day field direction, B ($D = -8°$, $I = 71°$)

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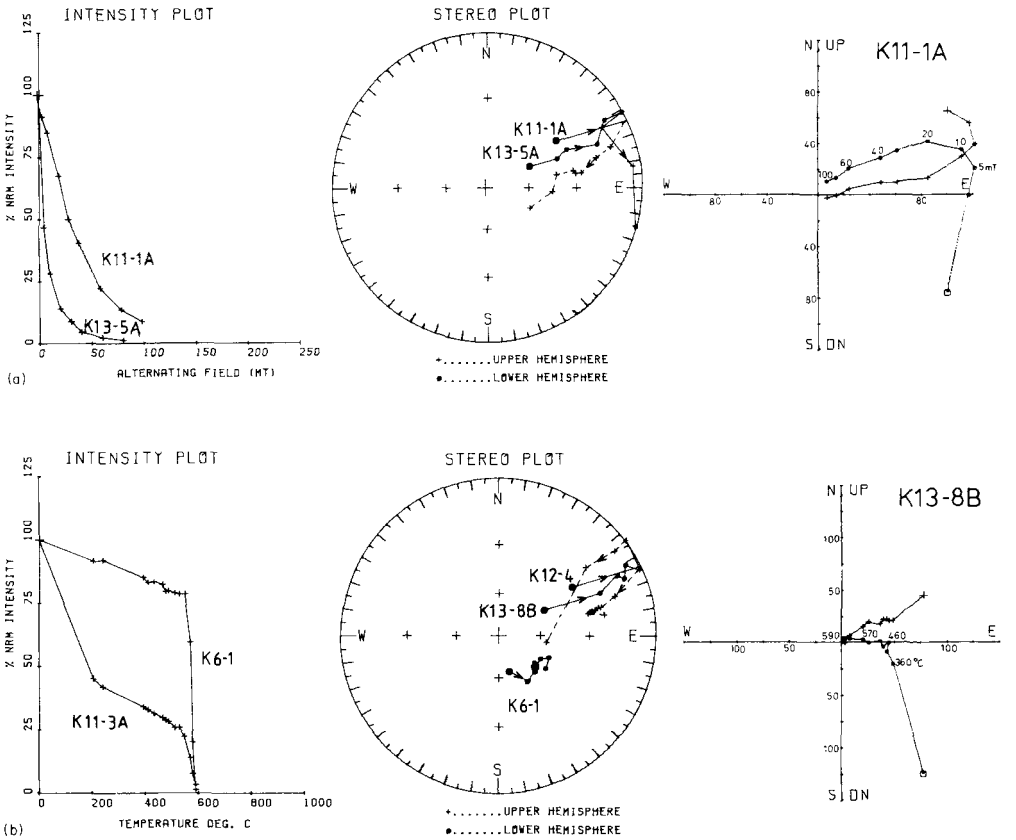



**Figure 7.** (a) Af and (b) thermal demagnetization behaviour of stable specimens from the Comrie diorite and hornfels (K6-1). On the orthogonal projections circles represent points on the vertical plane and crosses represent points on the horizontal plane. Axes are marked in units of $\times 10^{-3}$ A m$^{-1}$.

(Fig. 8). Therefore, it is assumed that, in these cases, overlap of blocking temperature ranges has prevented the complete removal of the present-day magnetization, B, from the primary component, A. The B component may be explained as a CRM due to post-cooling weathering processes.

A steep south-easterly, downward magnetization, C, was observed in the hornfels and in some specimens from site K17. Examples show that C has lower Tb but higher coercivity than A.

## 4.3 MAGNETIC PROPERTIES

Thermomagnetic curves of the diorites (e.g. Fig. 9a) have the shape characteristic of magnetite. Curie points between 574 and 580°C indicate that the composition is pure magnetite. The result for KK3-4 (544°C), indicates the presence of Ti in the composition (< 0.1 mole fraction of $Fe_2TiO_4$) which could explain the poor magnetic stability compared with members of the geographically close site KK2. There is no evidence in the shape of the curve to suggest the presence of superparamagnetic material; therefore the poor stability is attributed to grains with low blocking temperature and short relaxation time. The presence

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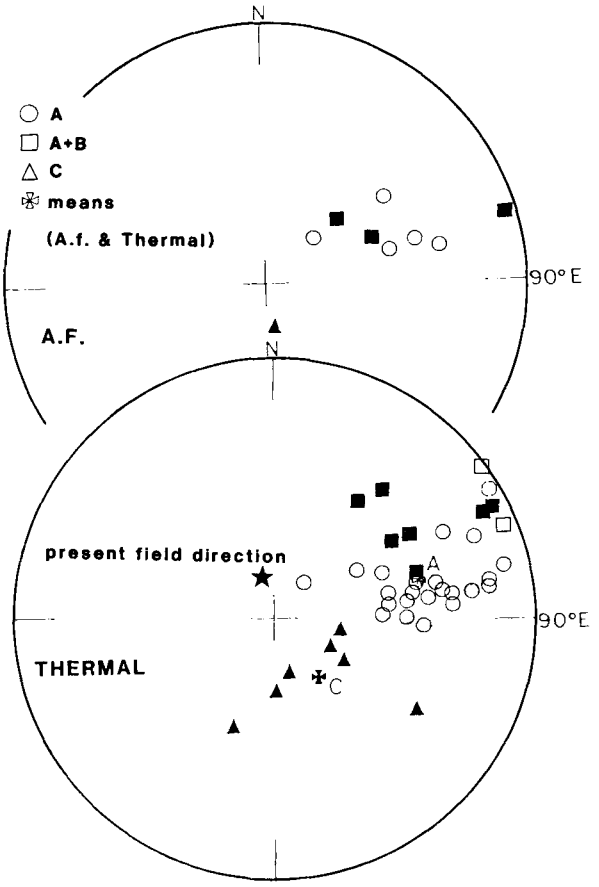



**Figure 8.** Stereographic plot for the Comrie intrusion illustrating the removed vectors after demagnetization. The classification of components A, B and C is explained in the text. Open symbols represent upper hemisphere directions, solid symbols represent lower hemisphere directions.

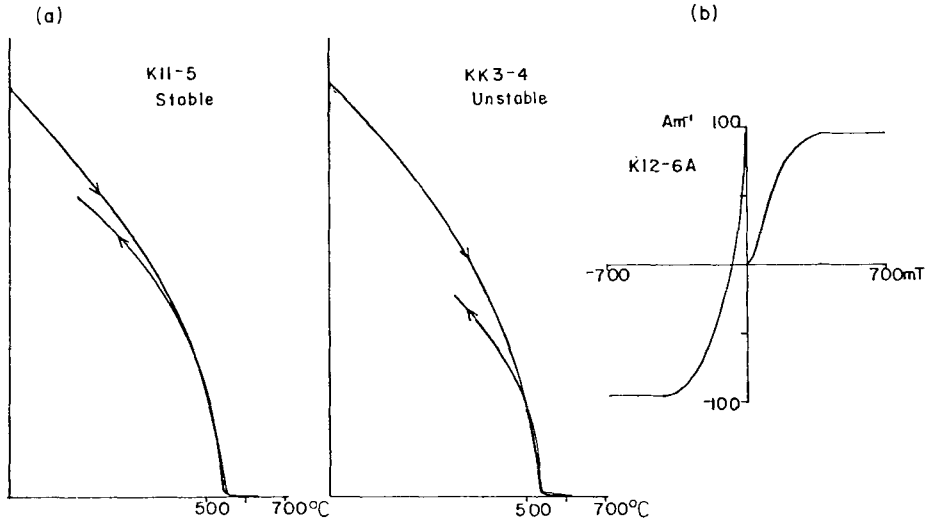



**Figure 9.** Typical (a) thermomagnetic and (b) saturation IRM curves from the Comrie collection. The vertical scale for the thermomagnetic curves, $J_S$, is arbitrary.

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of magnetite obscures the detection of haematite by the Curie balance because of the susceptibility contrast. However, it is clear from the low $B_{cr}$ values and saturation field strengths measured in the IRM experiments (e.g. Fig. 9b) that there is none present in any of the samples tested. The high saturation magnetization measured in all specimens is due to the grains larger than 100 $\mu$m in diameter observed in the microscope studies.

Thin-section studies revealed some clear differences in the character of stable and unstable rock-types: (i) the grain size of the stable material is generally finer; (ii) the stable rock is fresher, showing less evidence of sericitization in the feldspar or oxidation of magnetite to haematite; (iii) there is a higher proportion of single phase magnetite without ilmenite exsolution lamellae in the stable material; (iv) the stable rock types show a higher proportion of small grain sizes among the magnetites. It is likely that these differences are shared by grains of sub-optical dimensions in the rocks. The coarsest material probably lacks any sd remanence carriers (with grain diameter below 15 $\mu$m). Significantly, only the stable rocks contain magnetite whose size range extends to the limit of resolution of the microscope.

### 4.4 INTERPRETATION

The C component is not precisely defined. It is restricted to the hornfels and some diorite specimens in site K17 suggesting that magnetite grains of the appropriate relaxation time must be limited to these rock types. Unfortunately, this renders the hornfels useless for clarifying the tectonic interpretation of the primary diorite magnetization. The direction does not correspond closely to any British magnetization since the Devonian. The only specimen with both C and A components reveals that C has the lower blocking temperature although this does not necessarily imply that C represents a younger magnetization than A. Maybe C is a combination of the present field direction and a post-Devonian magnetization, perhaps Tertiary or Carboniferous.

Comparison of the characteristic direction of magnetization (A) with those derived from other Highland intrusions of comparable age, e.g. Garabal Hill-Glen Fyne and Arrochar (Briden 1970) reveals a considerable discrepancy in declination. However, a similar primary magnetization with easterly declination has been reported from the Cwmbran Old Red Sandstone (Chamalaun & Creer 1964) and, more recently, westerly palaeomagnetic poles have been derived from metamorphic rocks in the Moines north of the Great Glen, Scotland (Watts 1982). If these directions represent true Devonian magnetizations, the primary component, A, found in Comrie is not anomalous but confirms a large westerly swing in the APWP for Britain at the end of the Silurian.

Alternatively, the Comrie characteristic direction may be a primary TRM comparable to magnetizations in other Newer Granites, which has been affected by the post-cooling rotation of the intrusion. The geology of the country rock north of the fault is structurally complicated. The Dalradian grits are part of the inverted limb of a large recumbent fold, the Tay Nappe. Detailed mapping by several different workers has shown that, at different localities adjacent to the fault, the inverted grits and slates have suffered varying degrees of folding (Harris 1972; Harris, Bradbury & McGonigal 1976; Shackleton 1958). The Tay Nappe itself is related to the Grampian metamorphism which ended long before 408 Ma, but the later folds may well be the result of movement of the Highland Boundary Fault which, because of its proximity and strike, could also have caused the displacement of the Comrie palaeomagnetic pole. Bulk rotation of the intrusion about a horizontal axis (strike 050°) parallel to the fault during vertical, downward faulting would produce the appropriate rotation (60°). This requires that the intrusion lies in the core of an antiform structurally between the Aberfoyle synform to the SE and the synform at Callander. While this is com-

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patible with its geographical location, there is no clear structural evidence in the country grits to support such a hypothesis.

Rotation about a vertical axis which could transform the Comrie characteristic direction to that of Garabal Hill–Glen Fyne is discounted because of the consistency in regional strike over the area and the lack of suitable strike-slip faults along which such a rotation might be taken up.

## 5 Geochronological analysis

### 5.1 METHOD

Samples were collected both from palaeomagnetic sites and from other parts of each intrusion, notably the coarser phases which were not suitable for palaeomagnetic work, in order to cover the full range of petrographic variation. Care was taken to choose the least

**Table 1.** Rb-Sr results.

| SAMPLE | Min | Rb conc. (ppm) | Sr conc. (ppm) | 87Rb/86Sr atomic ratios | 87Sr/86Sr atomic ratios |
|---|---|---|---|---|---|
| RATAGAN | | | | | |
| 4572 | WR | 57.3 | 1919.7 | 0.0864 | 0.70613 +/- 2 |
| | fsp | 47.1 | 2409.0 | 0.0566 | 0.70648 +/- 3 |
| | bt | 375.7 | 46.93 | 23.47 | 0.84339 +/- 8 |
| 4575 | WR | 64.5 | 2224.8 | 0.0839 | 0.70607 +/- 5 |
| | fsp | 51.4 | 3065.6 | 0.0485 | 0.70587 +/- 2 |
| | bt | 522.2 | 117.6 | 12.94 | 0.78238 +/- 20 |
| 4578 | WR | 61.2 | 1887.0 | 0.0938 | 0.70673 +/- 2 |
| | fsp | 71.7 | 3781.7 | 0.0548 | 0.70651 +/- 6 |
| | bt | 271.2 | 81.46 | 9.686 | 0.76363 +/- 14 |
| 4580 | WR | 186.4 | 1904.0 | 0.2832 | 0.70731 +/- 8 |
| | fsp | 56.8 | 5003.0 | 0.0328 | 0.70600 +/- 10 |
| | bt | 451.5 | 90.20 | 14.60 | 0.79166 +/- 12 |
| COMRIE | | | | | |
| 4584 | WR | 33.8 | 829.9 | 0.1178 | 0.70560 +/- 3 |
| | fsp | 14.4 | 1539.2 | 0.0270 | 0.70504 +/- 2 |
| | bt | 398.4 | 39.13 | 29.95 | 0.87879 +/- 4 |
| 4585 | WR | 110.6 | 529.9 | 0.6038 | 0.71012 +/- 5 |
| | fsp | 42.6 | 751.0 | 0.1639 | 0.70750 +/- 3 |
| | bt | 556.2 | 21.60 | 77.59 | 1.13168 +/- 7 |
| 4589 | WR | 158.9 | 498.1 | 0.9232 | 0.71038 +/- 5 |
| | fsp | 91.3 | 428.6 | 0.6166 | 0.70868 +/- 2 |
| | bt | 715.1 | 21.98 | 99.41 | 1.28291 +/- 5 |

| Standard | Rb/Sr | for comparison: |
|---|---|---|
| NBS607 | 8.054 | |
| NBS607 | 8.060 | 8.00 +/– 0.05 *1 |
| GSP1 | 1.0905 | 1.0927 *2 |

*1 – US National Bureau of Standards specification.
*2 – Pankhurst & O'Nions (1973).
Mineral code: WR whole rock, fsp feldspar, bt biotite.

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altered specimens, identified from thin section studies. for Rb–Sr isotopic analysis. These were 4572, 4575, 4578 and 4580 from Ratagan and 4584, 4585 and 4589 from Comrie.

Standard sample preparation techniques were used. Rb was analysed using triple tantalum filaments on a VG-Micromass30 mass spectrometer. Sr was analysed on an automatic VG-Isomass54E mass spectrometer. Rb/Sr ratios are estimated accurate to 0.7 per cent. Analyses of standard samples are given in Table 1. SRM 987 analysed on the 54E during the period of study gave a mean of 0.710258 ± 6 (2 se, 76 analyses).

## 5.2 RESULTS

The results are presented in Tables 1 and 2. High values (> 1) of the mean square of weighted deviates (mswd) result from the failure of all points to lie on an isochron within error. This may be related to the sericitization of the feldspar, implying mobility of Sr near the feldspar grains. It is unlikely that Sr was mobile on the whole rock scale. Thus, where appropriate (e.g. 4572, 4580 and all the Comrie samples), ages have also been calculated from the biotite and whole rock data alone. This approach may be justified by the homogeneity in initial ratios of three of the Ratagan samples (Table 1). Sample 4585 from Comrie gives an anomalously young age. This may reflect the location of this sample in the contact zone between the diorite and the surrounding grits.

Calculations of the mean age for each complex give 415 ± 5 Myr for Ratagan and 408 ± 5 Myr for Comrie and probably dates the cooling of the intrusions below about 300°C. The error on the age difference between Ratagan and Comrie, 7 Myr, is probably less, say ± 2 Myr which is important when comparing the palaeomagnetic poles derived from the different intrusions.

**Table 2.** Age determinations.

| SAMPLE | AGE Ma | mswd | Initial 87Sr/86Sr atomic ratio |
|---|---|---|---|
| RATAGAN | | | |
| 4572 (bi-WR-fsp) | 408 +/- 138 | 611 | -- |
| 4572 (bi-WR) | 412.2 +/- 5.7 | -- | 0.70563 +/- 3 |
| 4575 (bi-WR-fsp) | 416.8 +/- 5.8 | 0.06 | 0.70558 +/- 2 |
| 4578 (bi-WR-fsp) | 416.5 +/- 5.8 | 0.13 | 0.70617 +/- 2 |
| 4580 (bi-WR-fsp) | 412.2 +/- 14.1 | 6.19 | -- |
| 4580 (bi-WR) | 413.7 +/- 5.8 | -- | 0.70564 +/- 9 |
| COMRIE | | | |
| 4584 (bi-WR-fsp) | 408.5 +/- 8.2 | 2.24 | -- |
| 4584 (bi-WR) | 407.7 +/- 5.6 | -- | 0.70492 +/- 2 |
| 4585 (bi-WR-fsp) | 390.7 +/- 24.7 | 25.64 | -- |
| 4585 (bi-WR) | 384.6 +/- 5.3 | -- | 0.70681 +/- 4 |
| 4589 (bi-WR-fsp) | 407.1 +/- 8.7 | 2.54 | -- |
| 4589 (bi-WR) | 408.2 +/- 5.6 | -- | 0.70501 +/- 7 |

Decay constant $\lambda_{Rb} = 1.42 \times 10^{-11}\,yr^{-1}$.

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**Table 3.** Palaeomagnetic south poles. For Ratagan the combined pole is calculated from site means using the mean normal components (found in seven sites) and the mean reversed components (found in five sites) as independent values. For the metasediments and Comrie, the within site variability resulted in too few stable components from some sites. Hence the pole is calculated from the specimen population. $S$ is the number of vectors used in the calculation, $R_S$ is the unit vector sum. $K$ is Fisher's (1953) precision estimate. $\alpha_{95}$ is the semi-angle of the cone of 95 per cent confidence about the mean direction. $\delta p$, $\delta m$ are the semi-axes of the oval of 95 per cent confidence about the mean pole.

| INTRUSION | LOCALITY Lat.N | Long.E | AGE Ma | DIRECTION D | I | S | K | $R_s$ | $\alpha_{95}$ | PALAEOMAGNETIC SOUTH POLE Lat.S | Long.E | $\delta p$ | $\delta m$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| RATAGAN,R | 57.2 | -5.5 | 415±5 | | | | | | | | | | |
| Normal | | | | 3.2 | -35.6 | 7 | 54.4 | 6.9 | 8.2 | -12.2 | 161.7 | | |
| Reversed | | | | 181.1 | 27.0 | 5 | 23.1 | 4.8 | 16.2 | 18.4 | 353.3 | | |
| *Combined* | | | | 8.0 | -32.2 | 12 | 30.4 | 11.6 | 7.9 | 15.0 | 346.5 | 5.0 | 8.9 |
| COMRIE,C (Nor.) | 56.4 | -4.0 | 408±5 | 75.1 | -30.3 | 28 | 19.2 | 26.6 | 6.3 | 6.2 | 287.1 | 3.9 | 7.0 |
| after rotation,C* | | | | 32 | -35 | | | | | 10 | 325 | | |
| METASEDIMENTS west of Ratagan,M (Rev.) | 57.2 | -5.6 | ? | 214.8 | 45.9 | 11 | 36.8 | 10.7 | 7.6 | 0.5 | 323.8 | 6.2 | 9.7 |

## 6 Conclusions

The palaeomagnetic poles and ages derived from the two intrusions at Ratagan and Comrie, together with results from the nearby metasediments, are listed in Table 3 and plotted in Fig. 10 together with the published APWP for Britain north of the Iapetus Suture for this

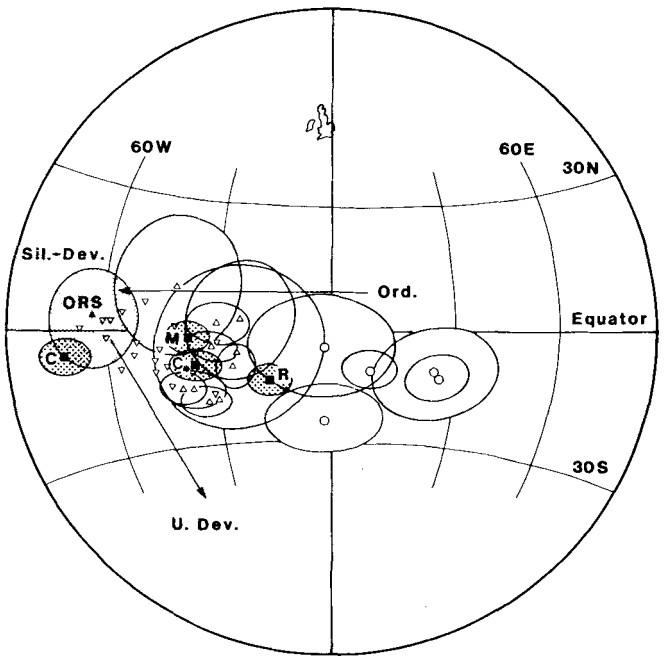



**Figure 10.** Palaeomagnetic poles of this study are shown as solid squares with darkly stippled confidence circles, and labelled as in Table 3. They are plotted against a background of previously published data; open symbols with associated circles of confidence represent data from north of the Iapetus suture. Circles ○ and triangles △ distinguish Ordovician and Siluro–Devonian formations respectively. Inverted triangles ▽ show site mean poles for the Moines, without confidence circles (for clarity) some of which lie as far west as the Cwmbran Old Red Sandstone pole and approach the position of the unrotated Comrie pole (after Turnell & Briden 1983).

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period (e.g. Turnell & Briden 1983). This shows that the Ratagan pole lies midway between the Ordovician and Siluro–Devonian groups and that the Comrie pole falls either (a) among the Newer Granite poles or (b) among the most westerly lower Devonian (?) poles after applying a post-cooling tectonic correction. Of the metasediments, only the reversed NRMs from west of Ratagan are adequate in number to provide a useful datum.

The youngest of the Ordovician poles (the Borrolan outer phase) is dated at 430 Ma (Van Breemen, Aftalion & Johnson 1979) while the Siluro–Devonian data north of the Southern Uplands, many of which have not been dated isotopically, range in age from *c.* 418 Ma (Arrochar) to *c.* 404 Ma (Garabal Hill–Glen Fyne). While the 415 Ma Rb/Sr data represent a lower limit for the age of the primary magnetization in Ratagan (with blocking temperature *c.* 580°C), comparison of Ratagan with neighbouring palaeomagnetic poles (Fig. 10) suggests that the ages are comparable, i.e. the intrusion cooled quickly. The spread of poles to the west of the Ratagan result indicates a period of rapid apparent polar wander after 415 Ma, including a large rotational component, which supports the hypothesis that the field reversal observed in the Ratagan results occurred during the initial cooling of the body. It is clear that the NRM of the metasediments west of Ratagan gives a pole far from any Ordovician data (i.e. in positions which might have been anticipated from the 430–480 Ma 'chrontour' ages (Dewey & Pankhurst 1970)). Therefore, it is assumed that the magnetization of these rocks is related to a younger event associated with prolonged Newer Granite activity. Similar ideas have been discussed in relation to palaeomagnetic results from the heavy mineral seams about 20 km south of Glenelg (Watts 1982) which also fail to agree with the published chrontour pattern.

The lack of reversed magnetization components among the Comrie results is unusual compared with the alkaline Newer Granites of the Assynt area (Turnell & Briden 1983) and Ratagan. It is assumed that, in this particular rock, evidence of reversed components with blocking temperatures lower than that of the assumed primary component, A, is completely obscured by the viscous overprint. The separation of the uncorrected Comrie pole from other Newer Granite poles of comparable age supports the idea that the intrusion has been affected by post-cooling rotation associated with the Highland Boundary fault. Against this hypothesis are (1) the lack of supportive evidence in the structure of the surrounding country grits and (2) the independent evidence, from the Welsh Old Red Sandstone and metamorphic rocks within the Moines, of a large westerly swing in the APWP for Britain in the late Silurian–early Devonian.

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